

Arithmetic Quantum Mechanics

Lab Book

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1 Overview

This campaign’s north star is a general definition and workflow that assigns to an arbitrary arithmetic scheme one or more *canonical quantum systems*: kinematics realised as a Hilbert space, observables realised as an algebra, and, where possible, dynamics realised as automorphisms or as projective unitary representations of symmetry groups, with fusion categories as the expected general endpoint. The word *canonical* is a technical requirement here, not a mood: the assignment must be exhibited as a functor, every choice on which it depends must be named explicitly, and a theorem must state exactly what is independent of those choices.

The concrete guiding path fixes a target before the generalities are attempted: the prime field $\mathbb{F}_p$ of characteristic p , its canonical symplectic space $\mathbb{F}_p \times \mathbb{F}_p$, and the finite Weyl–Heisenberg system quantizing it. Every general definition this campaign proposes must reproduce this case exactly, and is tested against it before anything else. The result is recorded in this lab book, one topic per section, each statement carrying its full hypotheses, an honest status, and its provenance (§1); nothing in these pages is asserted at a strength its evidence does not support.

Status

No claim has yet been established in this campaign. The argument DAG `claims/CLAIMS.md` carries its schema and its status vocabulary but no rows, and `definitions.md` carries the conventions governing it but no numbered definitions. The table below shows the columns this section will carry once the DAG is populated; its single row is a placeholder, not a claim.

Claim id	Statement	Status
<i>(none yet)</i>	<i>No claim has been established. This row records the table’s columns only; it is added to, one row per entry of <code>claims/CLAIMS.md</code>, as the argument DAG is populated.</i>	—